SALES DATA CALCULATOR

1. Introduction

The Sales Data Calculator is a Python-based project developed to analyze a collection of sales values and calculate important statistical measures. The program determines the total sales, average sales, highest sale, and lowest sale from a manually created dataset.

This project demonstrates how Python can simplify numerical data processing and automate basic sales calculations.

2. Objective

The main objective of this project is to understand basic numerical data analysis using Python.

The specific objectives are:

- To create and process a list of sales values.
- To calculate total sales using Python's built-in functions.
- To determine average sales.
- To identify the highest and lowest sales values.
- To understand the practical application of numerical data analysis.

3. Problem Statement

Businesses record multiple sales transactions. Manually calculating the total, average, maximum, and minimum sales values can be repetitive and may result in errors.

This project aims to automate these calculations using Python and display the results in a structured format.

4. Tools and Technologies Used

Python 3: Used to develop the sales analysis program.

Visual Studio Code: Used to write, edit, and execute Python code.

GitHub: Used to store the source code and project documentation.

5. Dataset Description

The project uses a manually created dataset containing ten fictional sales values.

The values are:

12000, 15000, 18000, 10000, 22000, 25000, 14000, 30000, 17000, 27000

The sales values are stored in a Python list and processed using built-in Python functions.

6. Methodology

The project follows these steps:

1. Create a list containing sales values.
2. Validate the data before performing calculations.
3. Calculate total sales using `sum()`.
4. Calculate average sales by dividing total sales by the number of records.
5. Identify the highest sale using `max()`.
6. Identify the lowest sale using `min()`.
7. Display the calculated statistics in a structured format.
8. Compare the results with the original sales data.

7. Sales Calculations

Total Sales

Total sales represents the sum of all sales values.

Formula:

Total Sales = Sum of all sales values

Average Sales

Average sales represents the mean value of the recorded sales transactions.

Formula:

Average Sales = Total Sales / Number of Sales

Highest Sale

The highest sale represents the largest sales value in the dataset.

Lowest Sale

The lowest sale represents the smallest sales value in the dataset.

8. Expected Results

For the sample dataset, the expected calculations are:

Metric	Expected Value
Number of Sales	10
Total Sales	₹1,90,000
Average Sales	₹19,000

| Highest Sale | ₹30,000 |
| Lowest Sale | ₹10,000 |

These values should be verified by running the Python program.

Screenshot Evidence: Insert a screenshot of the actual program output here.

9. Testing and Validation

The program should be tested using different sales datasets to verify its accuracy.

Testing scenarios include:

- Valid sales values.
- An empty sales list.
- Negative sales values.
- Non-numeric inputs.
- A single sales record.
- Repeated sales values.
- Zero sales values.

The actual results should be recorded after execution and compared with the expected outcomes.

10. Concepts Demonstrated

The project demonstrates the following Python concepts:

- Lists and variables
- Functions
- Loops
- Conditional statements
- Built-in numerical functions
- Input validation
- Exception handling
- Structured output

11. Limitations and Future Improvements

The current implementation analyzes a small, manually created dataset.

Future improvements could include:

- Reading sales data from CSV files.
- Analyzing sales by product or category.
- Creating charts and graphs.
- Exporting results to a file.
- Developing an interactive dashboard using Streamlit.

12. Conclusion

The Sales Data Calculator demonstrates how Python can be used to automate basic numerical data analysis. By applying built-in functions and simple programming logic, the program calculates useful sales statistics from a collection of values.

This project provides practical experience in Python programming and establishes a foundation for more advanced data analysis applications.

13. Project Links

GitHub Repository: https://github.com/venky1318/sales-data-calculator